

Estimation of Average Grain Size from Microstructure Image using Machine learning and python.



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Abstract: In this study, image processing and machine learning were used to assess the average grain size from a microstructure image. It is more suited to test the training results quantitatively because, unlike the chemical composition or mechanical properties of a material, the grain size in a microstructure image can be directly measured and validated in the original image. It is possible to manually analyze microstructure images, such as grain size, or use image analysis; however, it is anticipated that machine learning will make the analysis easier and faster. A phase-field simulation was used to make microstructure images, and machine learning was done with the help of the data (CSV file) that was obtained after the image processing.

Keywords: machine learning; python; image processing; microstructure; FESEM; OpenCV

1. Introduction

As a result of the direct relationship between a material's microstructure and its mechanical properties, material designers constantly monitor and regulate the microstructure as they create new materials in an effort to obtain certain mechanical properties. The strength, hardness, elongation, and fatigue properties of materials are very well known to be closely connected to the average grain size of the material. Inhibiting grain growth by precipitating a second phase has been the subject of numerous studies on refining grains. When examining a microstructure, the grain size is the most basic property that is examined. Grain size estimation is a quantitative analysis to determine the average grain size of a polycrystalline material .

There are various techniques to estimate the grain size , like,

1. Linear Intercept Method : On micrographs, random lines are drawn, and the number of times these lines cross grains is recorded. The average grain size can be determined by measuring the intercepted lines' lengths and using a mathematical calculation.
2. Area Fraction Method : By overlaying a grid on micrographs, this technique counts the number of grains that cross the gridlines. The average grain size can be calculated using the number of grains and the size of the grid.
3. Image processing : The development of digital image processing methods has led to the development of software systems that can analyze micrographs and determine grain sizes automatically. When compared to manual processes, these instruments may be quicker and more precise.

In this study, a simple OpenCV program was created , and the average grain size was estimated from microstructure images. It is very difficult to secure enough actual microstructure images , so, these images were obtained by proper cutting and polishing of an alloy.

2. Model Description

2.1. Preparation of Sample

A piece of P91 grade Steel was taken to take the microstructure images used for the average grain size analysis.

A chrome-moly alloy metal with exceptional strength and temperature resistance is P91. It is a creep strength enhanced ferritic (CSEF) since it is made for increased creep strength. To create this metal, it is normalized at 1050 °C, air cooled to 200 °C, and then heated to 760 °C to temper it. The primary cause of P91's creep strength and durability is this process. P91 denotes the metal's 9% chromium, 1% molybdenum, and vanadium makeup. The chromium boosts oxidation resistance and thermal strength. The addition of molybdenum improves the material's elasticity, wear resistance, and high temperature creep strength. A small junk of P91 was cut into 5 small cuboids which were then heat treated at temperatures 900 , 950 and so on up to 1100 °C respectively. The Normalizing process shows an effective impact on the microstructural constituents of the steel . Normalization is performed for 45 - 60 min at different temperatures mentioned above. The samples were then cut into half . The remaining halves of the samples were then tempered at a temperature of 760 °C to form the precipitate which would be analyzed under the FESEM, these remaining halves were termed as Tempered Samples(T) and the earlier were termed as Normalized Samples(N).

All the 10 samples were then polished using different grades of polishing paper from 80 to 2000 for nearly 15-20 mins per sample on every paper , later samples were polished on a polishing cloth for 20 min on a polishing machine.

The polished samples were then etched with Vilella and Nital to study the efficiency of the respective etchants on the grain boundaries. A 100 ml solution of Vilella was made with 95ml Ethyl Alcohol , 5 ml HCl and 1g Picric Acid , also a 100 ml Nital solution was also prepared with 90 ml Ethyl Alcohol and 10 ml Nitric Acid (HNO₃). Few samples were then etched with Vilella Only a few with Nital and few with the mixture of Vilella and Nital with varying time of the etching process like 45s , 50 s , 55s and 60s . After the etching the samples were studied to find optimal etchant and etchant condition which gave a proper grain boundary to study. Finally, after the analysis of the images for the various etchant condition it was concluded that Vilella for 60s for optimal grain boundary analysis. The tested samples were again repolished and then finally etched with Vilella (95% Ethyl Alcohol , 5% HCl and 1g Picric acid) for 60s.

Now, The samples were ready for capturing the microstructure images which would be used further for the analysis using the python Program.

2.2. Capturing Microstructure Images using Optical Microscope

The polished and ready samples (10 in number) were taken first to Optical Microscope . The microscope used was Lecia DM1750 M . Microstructures images were taken for all the samples under various magnifications of 20x , 50x and 100x at various locations on the samples.

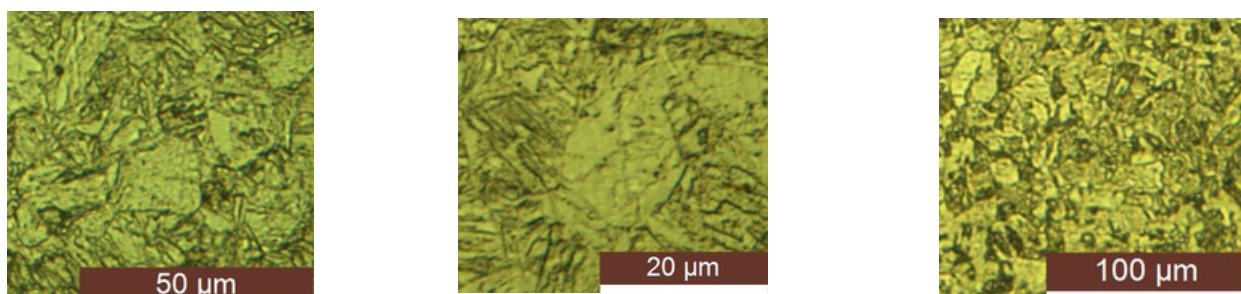


Figure 2. Schematic structure of various microstructure used in this study.

2.3. Capturing FESEM Images

After the Optical Images were taken , the samples were prepared for capturing the FESEM images . The samples must be conductive in order to capture high quality image without charging effects, so , the samples were then coated with a thin layer of gold using a sputtering machine. The samples were now ready to be captured under the FESEM , before loading the samples were numbered from 1 to 10 (1 to 5 were for N from 900 to 1100 °C resp. and 6 to 10 for T from 900 to 1100 °C resp.). Samples were then loaded into the sample holder , image parameters were then set to optimize the image quality. The electron beam was then positioned to capture images of the desired sites , the site was then focused on a particular magnification and then images were captured.

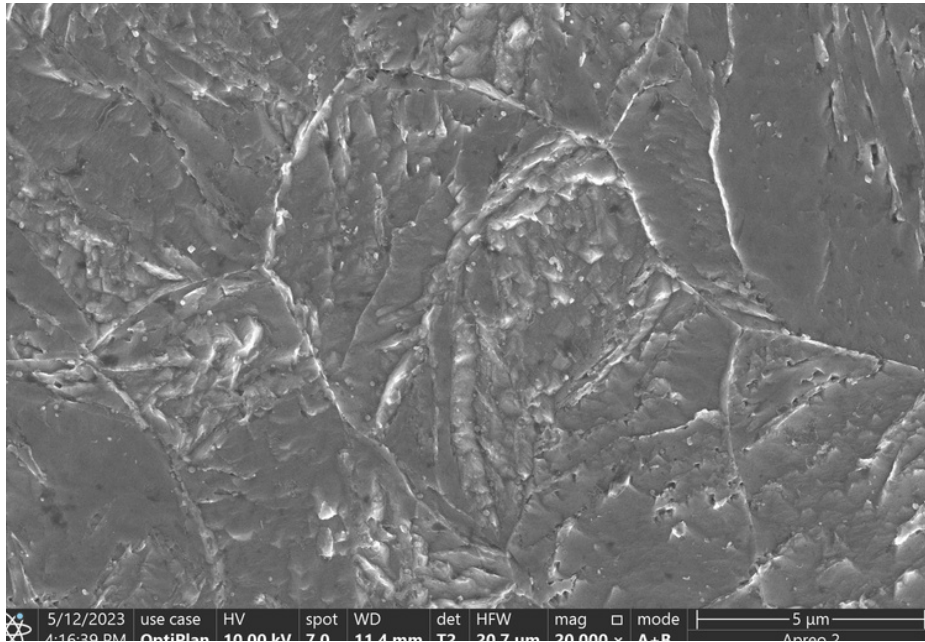


Figure 2.3a. FESEM image of N900

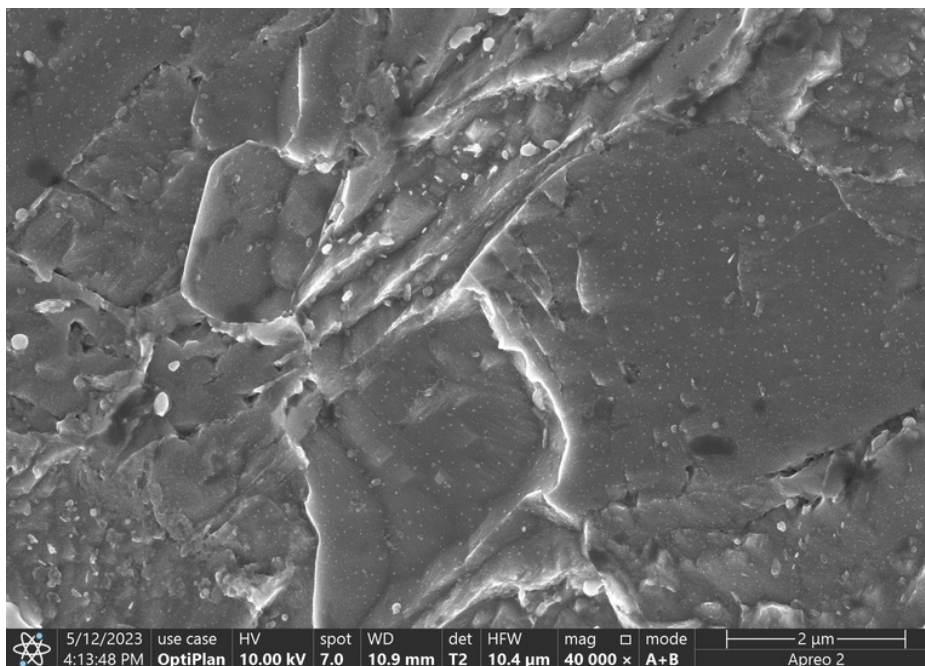


Figure 2.3b FESEM image of T900

2.4 Making of OpenCV Program

```
pip install opencv-python

In[ ]:

import numpy as np
import cv2
from matplotlib import pyplot as plt
from skimage import io,color,measure
from scipy import ndimage
img=cv2.imread(r"E:\\Second year\\SEM 2\\DC\\New-20230513T062323Z-001\\New\\T900\\100x1-000.tif")
cv2.imshow('image',img)

pixels_to_um=0.5
cropped_img=img[0:300,:]
```

Figure 2.3a Reading the image and defining the pixel size

```
ret,thresh=cv2.threshold(img,0,255,cv2.THRESH_BINARY+cv2.THRESH_OTSU)

# In[ ]:

kernel=np.ones((3,3),np.uint8)
eroded=cv2.erode(thresh,kernel,iterations=1)
dilated=cv2.dilate(eroded,kernel,iterations =1)
#cv2.imshow("Threshold Image",thresh)
#cv2.imshow("Dilated Image",dilated)
cv2.waitKey(0)

#converting to binary
mask=dilated==255
#io.imshow(mask[250:280,250:280])

#step4
s=[[1,1,1],[1,1,1],[1,1,1]]
labeled_mask,num_labels=ndimage.label(mask,structure=s)
img2=color.label2rgb(labeled_mask,bg_label=0)
cv2.imshow("coloured Labels",img2)
```

Figure 2.3b Denoising of the image

```
# In[ ]:

clusters=measure.regionprops(labeled_mask,img)

# In[ ]:

print(clusters[0].perimeter)

# In[ ]:

#step6
propList = ['Area',
            'equivalent_diameter', #Added... verify if it works
            'orientation', #Added, verify if it works. Angle btwn x-axis and major axis.
            'MajorAxisLength',
            'MinorAxisLength',
            'Perimeter',
            'MinIntensity',
            'MeanIntensity',
            'MaxIntensity']
```

Figure 2.3c Measuring the properties of each grain

```
# In[ ]:

output_file=open('image_measurements.csv','w')
output_file.write(','+'+'.join(propList)+'\n'))

# In[ ]:

for cluster_props in clusters:
    #output cluster properties to the excel file
    output_file.write(str(cluster_props['Label']))
    for i,prop in enumerate(propList):
        if(prop == 'Area'):
            to_print = cluster_props[prop]*pixels_to_um**2 #Convert pixel square to u
        elif(prop == 'orientation'):
            to_print = cluster_props[prop]*57.2958 #Convert to degrees from radians
        elif(prop.find('Intensity') < 0): # Any prop without Intensity in its
            to_print = cluster_props[prop]*pixels_to_um
        else:
            to_print = cluster_props[prop] #Remaining props, basically the ones with In
    output_file.write(',' + str(to_print))
output_file.write('\n')
```

Figure 2.3d Generating the csv file from the program

3. Working of the Program

1. In this programme, the images produced by the optical microscope are utilised. Image processing is the foundation of this programme.
2. This code makes use of libraries like matplotlib, skiimage, and scipy.
3. Scale of 0.5 is applied to convert picture pixels to micrometres.
4. The location of the pixel with the most grains is indicated by a histogram.
5. The binary threshold method is used to perform the image thresholding.
6. Additionally, erosion and dilation are carried out so that the grain structure's boundaries can be filled utilising the programme.
7. The regionprops function calculates statistics for each item (connected component) in an image, including area, centroid, and bounding box.
8. Finally, the program generates the necessary csv file and saves it to the system.
9. The data for machine learning is employed in the generated csv file, and output is produced.

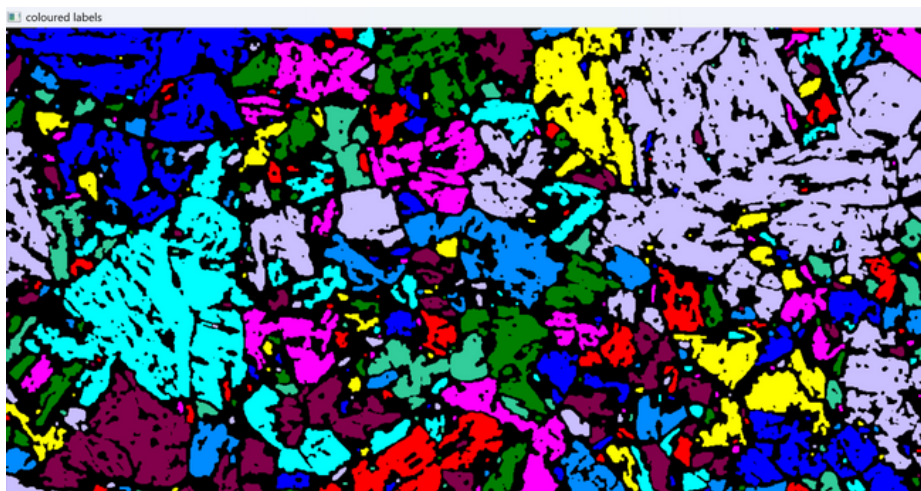


Figure 3.a coloured label image



Figure 3.b eroded image

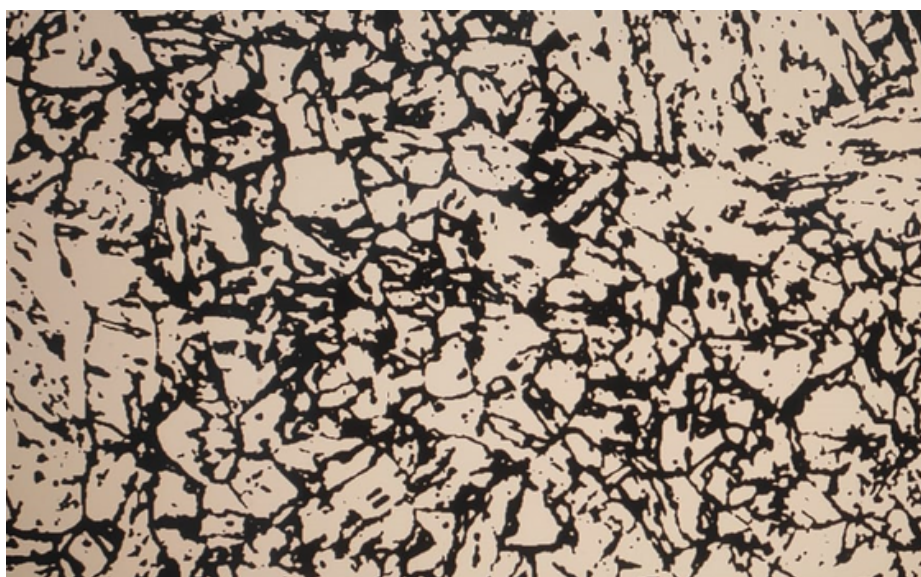


Figure 3.c threshold image

4. Results and Observation

4.1. Verifying the program Results

The microstructure images from the prepared samples earlier of (N and T) were then used for analyzing the efficiency of the results obtained from ImageJ (A popular image processing software used for finding average grain size) and from the python program. OpenCV program was used on the optical images and the ImageJ software was applied on the images obtained from the FESEM to verify if we obtain the same results or not. The images used was of the sample N900 (Normalized at 900 °C).

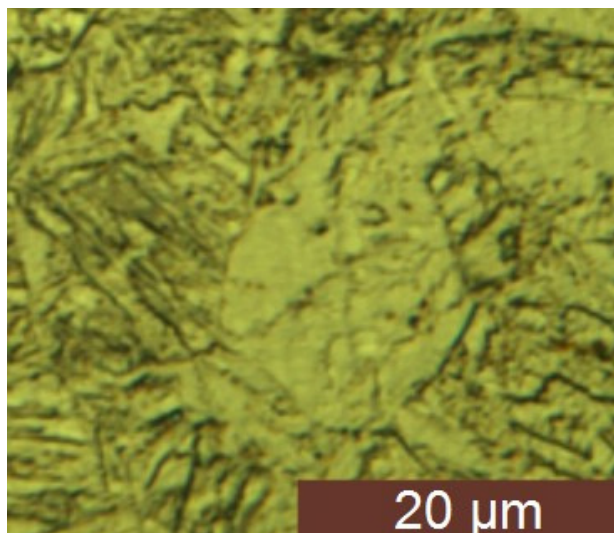


Figure 4.a N900 optical image

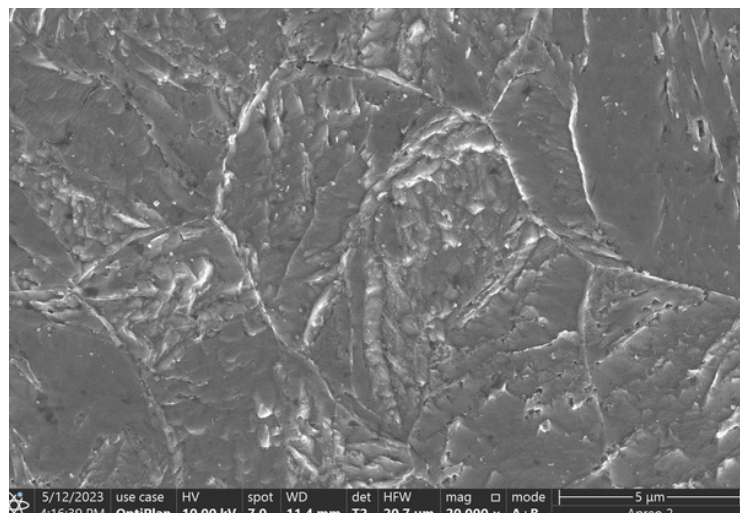


Figure 4.b.FESEM image of N900

```
In [1]: import numpy as np
import pandas as pd

In [2]: df=pd.read_csv(r"C:\Users\omiit\Downloads\image_measurements_100x_4.csv")

In [3]: df.mean(axis=0)

Out[3]: Unnamed: 0      480.500000
Area      185.058594
equivalent_diameter      6.056512
orientation      6.516607
MajorAxisLength      9.810819
MinorAxisLength      5.056262
Perimeter      81.046551
MinIntensity      132.660063
MeanIntensity      146.015949
MaxIntensity      161.787278
dtype: float64
```

Figure 4.a Avg. diameter obtained from csv file

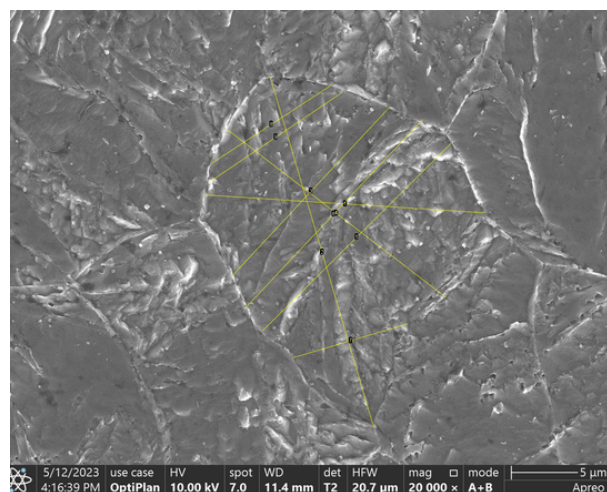


Figure 4.b Avg. diameter calculated using ImageJ

- The average diameter calculated from the csv file that the program generated is compared to the average diameter calculated from the grain size of the FESEM image.
- The avg. diameter obtained from the csv file is **6.05 micrometer** shown in the fig 4.a
- The avg. diameter of FESEM image is **6.137 micrometer** calculates using the ImageJ software as shown in fig 4.b

4.2. Variation of Temperature and Avg. Grain size

1. In materials science and metallurgy, the fluctuation of grain size with temperature is a key topic. The concept of recrystallization and grain growth can be used to generalize the relationship between grain size and temperature.
 2. **Recrystallization:** A metal's crystalline structure is altered when it undergoes plastic deformation (such as rolling, forging, or drawing). As the temperature rises, the energy within the deformed structure grows, and the substance tends to recrystallize to absorb some of this energy. New, strain-free grains nucleate and develop inside the distorted material during recrystallization. Generally speaking, higher temperatures encourage recrystallization, which leads to smaller grain sizes.
 3. **Grain growth:** On the other hand, if a material is kept at high temperatures for an extended period of time, the already-existing grains may expand in size. Grain growth is the name given to this process. It happens as a result of atoms diffusing over grain boundaries, causing nearby grains to consolidate. Larger grain sizes are often the result of grain growth, which may have an impact on the material's mechanical, thermal, and electrical properties.
1. **Higher temperatures:** In general, they encourage recrystallization and can cause strain-free grains to nucleate and develop in smaller grains.
 2. **Extended High-Temperature exposure:** As atoms diffuse across grain boundaries and adjacent grains combine, it may cause grain expansion and result in bigger grain diameters.

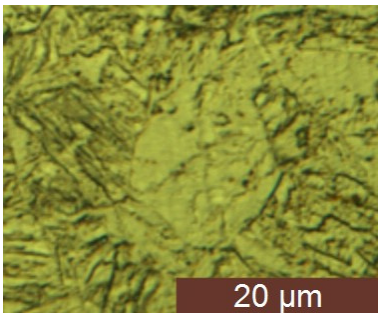


Figure 4.2a N900

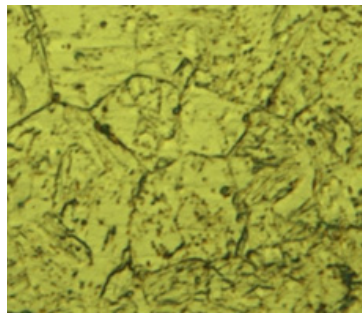


Figure 4.2b N950

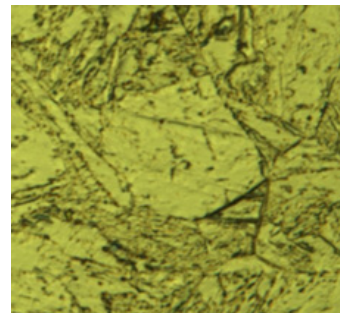


Figure 4.2c N1000

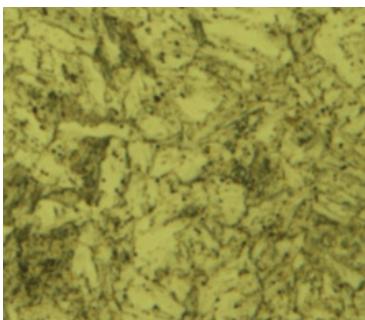


Figure 4.2d T900

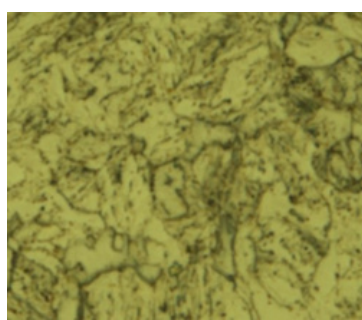


Figure 4.2e T950

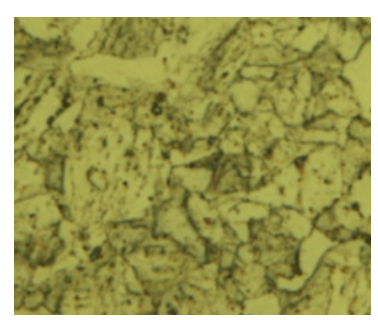


Figure 4.2e T1000

4.3 Machine learning

1. Machine also used to find the average grain size using various models on the csv file obtained from the program
2. The quality of the data, the selected features, and the suitability of the algorithm all affect how well the machine learning model performs.

5. Conclusions

In this study, the average grain size was evaluated from microstructure images using python and machine learning . . OpenCV gave the csv file which was then used to obtain the avg grain size and the same file could be used in a machine learning model to obtain results for same type of microstructure images. After all the tasks , the following conclusions were obtained:

- (1) The python code could be a better option for finding avg grain size rather than any other experimental procedures which nearly gave same results.
- (2) The CSV file obtained also gave other important parameters of study other than the grain size.
- (3) Machine learning models could be trained on the csv file data obtained and could be used to predict avg grain size of similar microstructure (optical) images.
- (4) Python program could save us the use of FESEM for the average grain size determination up to some extent.